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United States Patent 12413912
Kind Code B2
Date of Patent September 09, 2025
Inventor(s) Ren; Li et al.

Sound device

Abstract

The present disclosure discloses a sound device including a frame, a magnet system, and a first vibration system and a second vibration system. The magnet system includes a first inner yoke, a main magnet fixed to the first inner yoke, and a second inner yoke fixed to one side of the main magnet away from the first inner yoke; the first vibration system includes a first diaphragm, a first coil surrounding the second inner yoke, and a first FPC electrically connected with the first coil; the first coil includes a first corner portion; the first FPC includes a first fixation portion, a first welding portion welded to the first coil, and a first elastic portion; the first welding portion is located on one side of the first corner portion away from the second inner yoke. The sound device has better magnetic field performance and vibration performance.

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Appl. No.: **18/327006**

Filed: **May 31, 2023**

Prior Publication Data

Document Identifier	Publication Date
US 20240187798 A1	Jun. 06, 2024

Related U.S. Application Data

continuation parent-doc WO PCT/CN2022/136722 20221206 PENDING child-doc US 18327006

Publication Classification

Int. Cl.: **H04R9/06** (20060101); **H04R3/00** (20060101); **H04R7/04** (20060101); **H04R7/18** (20060101); **H04R9/02** (20060101); **H04R9/04** (20060101)

U.S. Cl.:

CPC **H04R9/063** (20130101); **H04R3/00** (20130101); **H04R7/04** (20130101); **H04R7/18** (20130101); **H04R9/025** (20130101); **H04R9/045** (20130101); H04R2400/11 (20130101)

Field of Classification Search

CPC: H04R (9/063); H04R (3/00); H04R (7/04); H04R (7/18); H04R (9/025); H04R (9/045); H04R (2400/11); H04R (2209/026); H04R (31/006)

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Background/Summary

FIELD OF THE PRESENT DISCLOSURE

(1) The present disclosure relates to electric-sound conversion technologies, especially relates to a sound device applied in portable mobile terminals.

DESCRIPTION OF RELATED ART

(2) With the development of the portable mobile terminal technologies, users have increasingly high requirements for the sound quality of portable mobile terminals. And a sound device in portable mobile terminals is one of the necessary conditions to achieve this high-quality sound effect

(3) In related art, the sound device normally includes a vibration system and a magnet system. The magnet system includes a yoke, a magnet fixed to the yoke, and a pole plate fixed to the magnet. The vibration system includes a diaphragm, a coil configured to drive the diaphragm, and a FPC

arranged between the diaphragm and the coil; a solder pad is provided on the FPC. Usually, the solder pad extends inside the coil and is located above the pole plate. For the convenience of welding, an avoidance groove corresponding to the solder pad is provided on the pole plate to avoid the solder pad, thus thinning the pole plate and resulting in loss of magnetic field BL. Besides, the glue configured to bond the FPC and the coil is prone to overflow to sides of the coil, negatively affecting the vibration ability of the vibration system.
(4) Therefore, it is necessary to provide an improved sound device to overcome the problems mentioned above.
SUMMARY OF THE INVENTION
(5) One object of the present disclosure is to provide a sound device having better magnetic field performance and vibration performance.
(6) A sound device includes a frame having a receiving space; a magnet system having a first magnetic gap and a second magnetic gap interconnected with each other, including: a first inner yoke; a main magnet fixed to the first inner yoke; and a second inner yoke fixed to one side of the main magnet away from the first inner yoke; a first vibration system arranged on one side of the magnet system, including: a first diaphragm fixed to the frame; a rectangular first coil inserted into the first magnetic gap and surrounding the main magnet and the second inner yoke; and a first FPC electrically connected with the first coil; a second vibration system arranged on the other side of the magnet system, comprising: a second diaphragm fixed to the frame; and a second coil inserted into the second magnetic gap and malposed with the first coil; the first coil includes a first corner portion located in corner position; the first FPC includes a first fixation portion fixed to the frame, a first welding portion welded to the first coil, and a first elastic portion connecting the first fixation portion with the first welding portion; the first elastic portion is spaced apart from the first coil; the first welding portion is located on one side of the first corner portion away from the second inner yoke.
(7) As an improvement, the first FPC includes two first welding portions; two welding portions are arranged along a diagonal of the first coil.
(8) As an improvement, the magnet system further includes a side magnet fixed to the frame, a first side yoke fixed to one side of the side magnet facing the first diaphragm, a second side yoke fixed to one side of the side magnet facing the second diaphragm, and a connection portion connecting the first inner yoke and the first side yoke; the first magnetic gap is formed between the first side yoke and the second inner yoke; the second magnetic gap is formed between the first inner yoke and the second side yoke; the first welding portion is malposed with the first side yoke along a vibration direction.
(9) As an improvement, a recessed portion opposite to the first welding portion is provided on the first side yoke along the vibration direction.
(10) As an improvement, a projection of the first coil along a vibration direction is located inside the second coil.
(11) As an improvement, the frame is in a rectangle shape; the side magnet is fixed to the frame along a long axis; the connection portion connects the first inner yoke with an edge of the first side yoke along a short axis.
(12) As an improvement, the second vibration system further includes a second FPC electrically connected with the second coil; the second FPC includes a second fixation portion fixed to one end of the second coil away from the second diaphragm, a second welding portion fixed to the frame, a second elastic portion connecting the second fixation portion with the second welding portion, and a third welding portion extending from the second fixation portion to outside of the second coil; the second coil is electrically connected with the third welding portion.
(13) As an improvement, the second vibration system further includes two second FPCs; two second FPCs are arranged on two opposite sides of the second coil along the short axis.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The present disclosure will hereinafter be described in detail with reference to an exemplary embodiment. To make the technical problems to be solved, technical solutions and beneficial effects of present disclosure more apparent, the present disclosure is described in further detail together with the figures and the embodiment. It should be understood the specific embodiment described hereby is only to explain this disclosure, not intended to limit this disclosure.

(2) FIG. **1** is an isometric view of a sound device in accordance with an exemplary embodiment of the present disclosure.

(3) FIG. **2** is an exploded view of the sound device in FIG. **1**.

(4) FIG. **3** is a cross-sectional view of the sound device taken along line A-A in FIG. **1**.

(5) FIG. **4** is an isometric view of part of the sound device in FIG. **1**.

(6) FIG. **5** is an isometric view of part of the sound device in FIG. **1**.

DETAILED DESCRIPTION OF THE EXEMPLARY EMBODIMENT

(7) The present disclosure will hereinafter be described in detail with reference to an exemplary embodiment. To make the technical problems to be solved, technical solutions and beneficial effects of the present disclosure more apparent, the present disclosure is described in further detail together with the figure and the embodiment. It should be understood the specific embodiment described hereby is only to explain the disclosure, not intended to limit the disclosure.

(8) Please refer to FIGS. **1-5**, a sound device **100** provided by an exemplary embodiment of the present disclosure includes a frame **1** having a receiving space **10**, a magnet system **2** received in the receiving space **10**, and a first vibration system **3** and a second vibration system **4** separately arranged on two sides of the magnet system **2**. The first vibration system **3** and the second vibration system **4** are separately mounted on two opposite sides of the frame **1**.

(9) The first vibration system **3** includes a first diaphragm **31** fixed to the frame **1**, a first coil **32** configured to drive the first diaphragm **31** to generate sound, and a first FPC **33** electrically connected with the first coil **32**.

(10) The second vibration system **4** includes a second diaphragm **41** fixed to the frame **1**, and a second coil **42** configured to drive the second diaphragm **41** to generate sound, and a second FPC **42** electrically connected with the second coil **42**. The second coil **42** is malposed with the first coil **32**. Specifically, a projection of the first coil **32** along a vibration direction is located inside the second coil **42**. It can be understood that the sound device **100** is in a rectangle shape. Correspondingly, the frame **1**, the first coil **32** and the second coil **42** are all rectangular.

(11) The magnet system **2** includes a first inner yoke **21**, a main magnet **22** fixed to the first inner yoke **21**, a second inner yoke **23** fixed to one side of the main magnet **22** away from the first inner yoke **21**, a side magnet **24** fixed to the frame **1**, a first side yoke **25** fixed to one side of the side magnet **24** facing the first diaphragm **31**, a second side yoke **26** fixed to one side of the side magnet **24** facing the second diaphragm **41**, and a connection portion **27** connecting the first inner yoke **21** and the first side yoke **25**. Therefore, a first magnetic gap **28** is formed between the first side yoke **25** and the second inner yoke **23**; a second magnetic gap **29** is formed between the first inner yoke **21** and the second side yoke **26**. The first magnetic gap **28** and the second magnetic gap **29** are interconnected with each other. And, the first coil **32** is inserted into the first magnetic gap **28**; the second coil **42** is inserted to the second magnetic gap **29**. Furthermore, the side magnet **24** is fixed to the frame **1** along a long axis; the connection portion **27** connects the first inner yoke **21** with an edge of the first side yoke **25** along a short axis.

(12) Specifically, the first coil **32** includes a first corner portion **321** located in corner position; the first FPC **33** includes a first fixation portion **331** sandwiched between the frame **1** and the first diaphragm **31**, a first welding portion **332** welded to the first coil **32**, and a first elastic portion **333**

connecting the first fixation portion **331** with the first welding portion **332**; the first elastic portion **333** is spaced apart from the first coil **32**; the first welding portion **332** is located on one side of the first corner portion **321** away from the second inner yoke **23**. In another words, the first welding portion **332** is located outside the first corner portion **321**. Owing that the second inner yoke **23** has significant impact on the BL of the magnet system **2**, the first welding part **332** is arranged outside the first corner portion **321** of the first coil **32**, rather than thinning the second inner yoke **23** to avoid the first welding part **332**. In this manner, the thickness of the second inner yoke **23** is thick enough to effectively improve the magnetic field performance of the magnet system **2**. Besides, the first elastic portion **333** is located outside the first coil **32**, not to be glued to the first coil **32**, thus effectively solving the glue overflow problem and improving the vibration performance of the first vibration system **3**.

(13) Furthermore, the first FPC **33** includes two first welding portions **332** two welding portions **332** are arranged along a diagonal of the first coil **32**. It should be noted that the two welding portion **332** are both located outside of its corresponding first corner portion **321**. It can be seen that the first welding portion **332** is malposed with the first side yoke **25** along the vibration direction. Specifically, a recessed portion **251** opposite to the first welding portion **221** is provided on the first side yoke **25** along the vibration direction. By providing the recessed portion **251** on the first side yoke **25** having less impact on the magnet system **2**, not only ensure the magnetic field performance of the magnet system **2**, but also make it more convenient to weld the first welding portion **332** and the first coil **32**.

(14) In addition, the second FPC **43** includes a second fixation portion **431** fixed to one end of the second coil **42** away from the second diaphragm **41**, a second welding portion **432** fixed to the frame **1**, a second elastic portion **433** connecting the second fixation portion **431** with the second welding portion **432**, and a third welding portion **434** extending from the second fixation portion **431** to outside of the second coil **42**; the second coil **42** is electrically connected with the third welding portion **434**. Specifically, the second vibration system **4** further includes two second FPCs **43**; two second FPCs **43** are arranged on two opposite sides of the second coil **43** along the short axis

(15) In one embodiment, as shown in FIGS. **2-3**, the sound device **100** further includes a front cover **5** above the second diaphragm **41** of the second vibration system **4**. The front cover **5** is space apart from the first diaphragm **41**. It can be understood that a sound hole **51** is provided on the front cover **5**. The sound generated by the second diaphragm **41** is transmitted to outside through the sound hole **51**. As shown in FIG. **5**, the third welding portion **432** of the second FPC **43** is electrically connected with external circuit through a conductive terminal **6**. The sound device further includes a conductive member **7** embedded in the frame **1**. The first FPC **44** is electrically connected with external circuit through the conductive member **7**.

(16) Compared with the related art, the sound device in present disclosure includes a frame, a magnet system, and a first vibration system and a second vibration system separately arranged on two sides of the magnet system. The magnet system includes a first inner yoke, a main magnet fixed to the first inner yoke, and a second inner yoke fixed to one side of the main magnet away from the first inner yoke; the first vibration system includes a first diaphragm, a first coil surrounding the second inner yoke, and a first FPC electrically connected with the first coil; the first coil includes a first corner portion located in corner position; the first FPC includes a first fixation portion fixed to the frame, a first welding portion welded to the first coil, and a first elastic portion connecting the first fixation portion with the first welding portion; the first elastic portion is spaced apart from the first coil; the first welding portion is located on one side of the first corner portion away from the second inner yoke. By arranging the first welding part outside the first corner portion of the first coil, rather than thinning the second inner yoke to avoid the first welding part, ensure the thickness of the second inner yoke, thus effectively improving the magnetic field performance of the magnet system. Besides, the first elastic portion is not glued to the first coil,

thereby effectively solving the glue overflow problem and improving the vibration performance of the first vibration system.

(17) It is to be understood, however, that even though numerous characteristics and advantages of the present exemplary embodiments have been set forth in the foregoing description, together with details of the structures and functions of the embodiments, the disclosure is illustrative only, and changes may be made in detail, especially in matters of shape, size, and arrangement of parts within the principles of the invention to the full extent indicated by the broad general meaning of the terms where the appended claims are expressed.

Claims

1. A sound device comprising: a frame having a receiving space; a magnet system having a first magnetic gap and a second magnetic gap interconnected with each other, comprising: a first inner yoke; a main magnet fixed to the first inner yoke; and a second inner yoke fixed to one side of the main magnet away from the first inner yoke; a first vibration system arranged on one side of the magnet system, comprising: a first diaphragm fixed to the frame; a rectangular first coil inserted into the first magnetic gap and surrounding the main magnet and the second inner yoke; and a first flexible printed circuit (FPC) electrically connected with the first coil; a second vibration system arranged on the other side of the magnet system, comprising: a second diaphragm fixed to the frame; and a second coil inserted into the second magnetic gap and malposed with the first coil; wherein the first coil comprises a first corner portion located in corner position; the first FPC comprises a first fixation portion fixed to the frame, a first welding portion welded to the first coil, and a first elastic portion connecting the first fixation portion with the first welding portion; the first elastic portion is spaced apart from the first coil; the first welding portion is located on one side of the first corner portion away from the second inner yoke; the magnet system further comprises a side magnet fixed to the frame, a first side yoke fixed to one side of the side magnet facing the first diaphragm, a second side yoke fixed to one side of the side magnet facing the second diaphragm, and a connection portion connecting the first inner yoke and the first side yoke; the first magnetic gap is formed between the first side yoke and the second inner yoke; the second magnetic gap is formed between the first inner yoke and the second side yoke; the first welding portion is malposed with the first side yoke along a vibration direction.
2. The sound device as described in claim 1, wherein the first FPC comprises two first welding portions; two welding portions are arranged along a diagonal of the first coil.
3. The sound device as described in claim 1, wherein a recessed portion opposite to the first welding portion is provided on the first side yoke along the vibration direction.
4. The sound device as described in claim 1, wherein a projection of the first coil along a vibration direction is located inside the second coil.
5. The sound device as described in claim 1, wherein the frame is in a rectangle shape; the side magnet is fixed to the frame along a long axis; the connection portion connects the first inner yoke with an edge of the first side yoke along a short axis.
6. The sound device as described in claim 5, wherein the second vibration system further comprises a second FPC electrically connected with the second coil; the second FPC comprises a second fixation portion fixed to one end of the second coil away from the second diaphragm, a second welding portion fixed to the frame, a second elastic portion connecting the second fixation portion with the second welding portion, and a third welding portion extending from the second fixation portion to outside of the second coil; the second coil is electrically connected with the third welding portion.
7. The sound device as described in claim 6, wherein the second vibration system further comprises

two second FPCs; two second FPCs are arranged on two opposite sides of the second coil along the short axis.